6) Experiment to understand feature scaling. Description: Understand the importance of feature scaling

Program:

```
import numpy as np
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
df=pd.DataFrame({

'Country':['USA','France','USA','China','France','USA','China',np.nan,'France','USA'],
    'Age':[22,34,27,np.nan,45,38,31,50,29,41],
    'Salary':[35000,60000,48000,72000,np.nan,55000,68000,80000,52000,64000],
    'Purchased':['No','Yes','No','Yes','No','Yes','No','Yes','No','Yes']
})
print("Original Dataset:")
print(df)
print("\n\nDataFrame Information:")
print(df.info())
df['Country']=df['Country'].fillna(df['Country'].mode()[0])
features=df.iloc[:,:-1].values
label=df.iloc[:,-1].values
age=SimpleImputer(strategy="mean",missing_values=np.nan)
Salary=SimpleImputer(strategy="mean",missing_values=np.nan)
age.fit(features[:,[1]])
Salary.fit(features[:,[2]])
features[:,[1]]=age.transform(features[:,[1]])
```

```
features[:,[2]]=Salary.transform(features[:,[2]])
print("\n\nAfter Filling Missing Values:")
print(features)
oh=OneHotEncoder(sparse_output=False)
Country=oh.fit_transform(features[:,[0]])
print("\n\nOne Hot Encoding:")
print(Country)
final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
print("\n\nFinal Set:")
print(final_set)
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
print("\n\nStandard Scaler:")
print(feat_standard_scaler)
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
print("\n\nMin Max Scaler:")
print(feat_minmax_scaler)
```

Output:

```
Original Dataset:
  Country   Age   Salary Purchased
0     USA  22.0  35000.0        No
1  France  34.0  60000.0       Yes
2     USA  27.0  48000.0        No
3   China   NaN  72000.0       Yes
4  France  45.0      NaN        No
5     USA  38.0  55000.0       Yes
6   China  31.0  68000.0        No
7     NaN  50.0  80000.0       Yes
8  France  29.0  52000.0        No
9     USA  41.0  64000.0       Yes
```

```
DataFrame Information:
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    9 non-null      str
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     str
dtypes: float64(2), str(2)
memory usage: 312.0 bytes
None
```

```
After Filling Missing Values:
[['USA' 22.0 35000.0]
 ['France' 34.0 60000.0]
 ['USA' 27.0 48000.0]
 ['China' 35.22222222222222 72000.0]
 ['France' 45.0 59333.333333333336]
 ['USA' 38.0 55000.0]
 ['China' 31.0 68000.0]
 ['USA' 50.0 80000.0]
 ['France' 29.0 52000.0]
 ['USA' 41.0 64000.0]]
```

```
One Hot Encoding:
[[0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]
 [1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]
 [1. 0. 0.]
 [0. 0. 1.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
Final Set:
[[0.0 0.0 1.0 22.0 35000.0]
 [0.0 1.0 0.0 34.0 60000.0]
 [0.0 0.0 1.0 27.0 48000.0]
 [1.0 0.0 0.0 35.22222222222222 72000.0]
 [0.0 1.0 0.0 45.0 59333.333333333336]
 [0.0 0.0 1.0 38.0 55000.0]
 [1.0 0.0 0.0 31.0 68000.0]
 [0.0 0.0 1.0 50.0 80000.0]
 [0.0 1.0 0.0 29.0 52000.0]
 [0.0 0.0 1.0 41.0 64000.0]]
```

```
Standard Scaler:
[[-5.00000000e-01 -6.54653671e-01  1.00000000e+00 -1.63305082e+00
  -2.00154053e+00]
 [-5.00000000e-01  1.52752523e+00 -1.00000000e+00 -1.50954278e-01
   5.48367268e-02]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00 -1.01551060e+00
  -9.32224356e-01]
 [ 2.00000000e+00 -6.54653671e-01 -1.00000000e+00  0.00000000e+00
   1.04189781e+00]
 [-5.00000000e-01  1.52752523e+00 -1.00000000e+00  1.20763422e+00
   5.98484550e-16]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00  3.43077904e-01
  -3.56438724e-01]
 [ 2.00000000e+00 -6.54653671e-01 -1.00000000e+00 -5.21478414e-01
   7.12877448e-01]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00  1.82517445e+00
   1.69993853e+00]
 [-5.00000000e-01  1.52752523e+00 -1.00000000e+00 -7.68494505e-01
  -6.03203995e-01]
 [-5.00000000e-01 -6.54653671e-01  1.00000000e+00  7.13602040e-01
   3.83857088e-01]]
```

```
Min Max Scaler:
[[0.         0.         1.         0.         0.        ]
 [0.         1.         0.         0.42857143 0.55555556]
 [0.         0.         1.         0.17857143 0.28888889]
 [1.         0.         0.         0.47222222 0.82222222]
 [0.         1.         0.         0.82142857 0.54074074]
 [0.         0.         1.         0.57142857 0.44444444]
 [1.         0.         0.         0.32142857 0.73333333]
 [0.         0.         1.         1.         1.        ]
 [0.         1.         0.         0.25       0.37777778]
 [0.         0.         1.         0.67857143 0.64444444]]
```

Result:

Experiments to understand the Feature Scaling is conducted.